

ACADEMIC QUALIFICATIONS

Year	Degree	Institute	Performance	
2021	B.Tech - Computer Science Engineering	Institute of Technical Education & Research, Bhubaneswar	7.77/10	
2016	Class XII (ISC)	Loyola School, Jamshedpur	72%	
2014	Class X (ICSE)	Loyola School, Jamshedpur	84%	

WORK EXPERIENCE

ITC INFOTECH	Full Stack GenAI Developer	Data & AI, Sept 2025 - Present
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NAMI Ask AI, External Client: Nippon India Mutual Fund

- Led the design and development of an internal AI assistant for Nippon India Mutual Fund Finance, enabling employees across departments (equity, banking, finance, marketing, digital, etc.) to create chat sessions, upload files, organize project workspaces, and search/query documents with contextual understanding.
- Used technologies such as LangChain, LlamaIndex, Python, FastAPI, React JS, NextJS, Docker, EC2, Lambda, ECS, S3, DynamoDB, DocumentDB, ElastiCache (Redis), Kinesis, API Gateway, VPC, Route 53, Transit Gateway, IAM, KMS, Secrets Manager.
- Built a full-stack GenAI platform with multi-session chat, project-based workspaces, and department-segmented file uploads using LlamaIndex agentic and fast models with an agentic tool-based retrieval approach.
- Developed the frontend using React + NextJS, delivering a clean, responsive chat UX with session history and workspace navigation on web, and used Flutter for Android and iOS applications.
- Integrated personalized RSS feeds and live news-based chat experiences tied to user registration preferences and external content sources.
- Delivered admin tools to track prompt usage, token metrics, and user activity, ensuring governance and transparency.
- Architected secure, scalable AWS infrastructure using serverless, containerized, and streaming services.

PRICEWATERHOUSECOOPERS US

Independent Audit Administrator,
Commercial Assets Technology & Associate
Solution Architect, Data Scientist & Full
Stack Developer

P&T Digital, Global IFS, Sept 2021 - May 2025

Independent Audit Administrator, Foreign Services

- Audited and managed foreign transactional services for private entities.
- Used technologies such as .NET, NodeJS, React, and Azure SQL services to manage global foreign policy MSA portfolios.
- Used .NET Core Entity Framework for API CRUD operations and LINQ queries for database migration to SQL Server, handling change requests and service requests from the client.

PwC Learning & Developments: PwC Full Stack Programme Boot Camps & PwC Certifications

- Achieved certifications for all PwC Full Stack Programmes, .NET, Java & Spring Framework, NodeJS, React & Angular Boot Camps during availability due to personal growth in learning & development field.
- Also got certified for Excellent quality Developer Persona Badge & internal training programmers apart from technology related stack such as for leadership qualities, learning pathways for individual health care as well as Gen AI 2.0.

External Client: Lupin Healthcare

- Solution Architect, Data scientist and Full stack developer role for one of the largest healthcare diagnostics center in India in application discovery phase using Python, Langchain, Elastic Search, Docker Networking, Azure, React, Ollama, Llama, & embedding models such as mxbai-embed-large as the most optimal solution technologies amongst lot others.
- Created complete backend for app discovery phase using above technologies as a proof of concept for implementation phase to generate response based on the form request parameters which generated required feature work item outputs.
- Created the entire end to end delivery for backend from solution architecture in app discovery phase, along with design and implementation along with project management and support from the external client.
- Created complete frontend to support the architecture with VanillaJS & React to encapsulate the above input forms for respective backend work items.

Microsoft Azure Devboards Product Release Notes Plugin

- Full stack developer role for Microsoft Azure Devboards Product Management, Release Notes Plugin using Semantic Kernel, Python, Azure, React, OpenAPI as the tech stack.
- Created complete backend for release 1.0 using skills and contexts where the semantic function was used to generate response based on the form request parameters which generated release notes.
- Above was done using creating prompt logic with planners to do different semantic tasks and give out a text, pdf or an html release notes document as output which was a downloadable buffer stream.
- Created complete frontend to support the architecture with React to encapsulate the above input form which populated custom drop-down with epic features with selection check boxes for release notes generation. The React frontend also provided handle form validation, generation requirements and preview components before downloading the reports.

Concourse 2.0

- Full stack developer & Data/Solution Design Architect role for Concourse 2.0.
- Played a key role in architecture design, migrate, implement, test, release from Monolith to Event Based Module Federated Sub Module Distributed Micro-Front Ends impacting organization's ROI by a huge profitable margin.
- Was a very core member for the entire product life cycle & took important decisions around all the core feature, user stories along the entire CI & CD pipelines using MongoDB, AzureServiceBus, RxJS, NestJS as backend & React & Angular as frontend technology.
- Also implemented various useful and logical distributed test designs using Unit, Integration, Performance & Regression testing using technology like Jest and SwaggerUI for the respective context.

Absolute Systems & Functions: PwC Digital-IFS

- Worked as Full stack developer as well as Data Engineer, Design & Architect role for ABS SF.
- Played a key role in activation of the ABS Master Data set & Inventory to support Performance Reporting.
 - Designed data entity relationship architecture and control flow chart which involved aligning on business requirements for interim & future states.
 - Built key mappings ABS Asset Names and Asset IDs using Excel formulas & script by writing cosine similarity for intensive fuzzy string matching.
 - Implemented mappings to other Inventory Governance data, Risk & Controls data & LeanIX field attributes with ABS data in Excel.
 - Used Power Query M for data ETL into Dataverse for utilization metric reporting.
- Developed PowerBI dashboards enabling Usage & Return on Investment Reporting dashboard, Operations Dashboard for Asset Management.
- Collaborated in development of Investment Intake Management Tool & Commercial Transparency Report for firm wide asset adoption & intake.
- Designed glossary related terms for entire ABS data model in Excel

AI Factory: PwC Digital Generative AI Initiative

- Worked as backend developer for a key Generative-AI internal service, **Chat-PwC**, with an **overall high impact and success margin, adoption rate and utilization value** for the firm.
- Created **Proof of concept of end to end application development and deployment** from given user story using ChatGPT from OpenAI.
 - Created a sequence of classes using **Structural Design Pattern**.
 - Base classes in sequence include: Deployment to **Databricks configuration class**, **OpenAI Chat-GPT Large-Language-Model**, class to engineer **Prompt Templates**. **OpenAI API Res ponse class** to generate response.
 - Only **1 single object** is created from the above Response class & used for all the **service types including back-end, front-end, unit test, docker & pipeline with added code comments**. **Configuration** for used technolgy is based on user preferences.
 - From the above OpenAI model **Code generation, Code optimisation & Unit test is generated for both back-end API and front-end API** used to create the **Requirement packages file** for both back-end & front-end code.
 - Finally, using **Chat-GPT**, a **Docker-file** was created for the end to end application & which was then used to make an **Azure pipeline** for deployment.
- Created a usecase for generative AI as a enhancer for **Project Planning Phase of Engagement Risk, Resolve, Receive, Plan, Create and Deliver Lifecycle**.
- Implemented **SKLearn & LLMs in reusable components as MFEs** to be used in throughout the PwC Digital department.

Concourse User Story Coach: PwC Digital-IFS

- **Optimization of recommendation hierarchy and capability importing API's** for given user stories from **18 to 12 seconds & 19 to 12 seconds** respectively using **Python, Flask & MongoAlchemy** with proper security including **JWT-Authentication** following best practices.
- **Implemented the whole unit test case module using Pytest** having least dependencies using concepts like **fixtures, test-client, scopes, conftest, config files & class testing**. Got the coverage results from **0 to 80 %**.
- Fixed codesmells with **Sonarqube** for successfull deployment **passing the Veracode scans**.
- **Upgraded Python from 3.8 to 3.9** which also required me to **change built in system packages from JWT due to version conflicts**.
- Fixed **Azure pipeline issue of coverage synchronization for backend Sonarqube and fixed other Azure pipeline issues**.

Insights Platform & Usage Tracker: PwC Digital Labs

- Leveraged **Builder Design Pattern** to write a package in **Python** that gets user information from **Reusable Accessible Components** containers on servers
- Connected the package to **MySQL** server with proper implementation of **JDBC** that retains user's information per session during data extraction
- Used **PySpark** to create data-frame from the user class object & converted it to **Pandas** data-frame for final upload to the server

Microsoft PowerApps Professional Developer Bootcamp, PwC L&D

- Designed & built **PowerApps** prototype application called **Device Ordering App** on **low code** environment which compares & selects laptops from store
- Inculcated **Dataverse & Power Component Framework** to design **Dynamic Text Input** fragment that stores order from the user & submits it to the supplier
- Built **Supplier Portal** using **Power Portals** that checks the users order & approves it based on the desired conditions applied from **Flow Automations**

PRICEWATERHOUSECOOPERS BADGES

- **P&T Digital: NodeJS Badge**
- **P&T Digital: Angular Badge**
- **P&T AWS Cloud Program Badge**
- **P&T Digital: PwC Quality Dev Persona**
- **P&T Digital Acumen**
- **P&T Digital: .NET Badge**
- **P&T Digital: Spring Boot Badge**
- **P&T Vantage: Developer Badge**
- **P&T Digital: Python Badge**
- **P&T Digital: Pro Dev**
- **P&T Digital: Human Centered Design**

INSTRUMENTATION & AUTOMATION DIVISION, TATA STEEL JAMSHEDPUR

Research Internship

May 2018 - June 2018

Pseudo defect reduction in Surface Inspection System at CRM Bara

Project Supervisor: Mr. Ashish Tiwari, Principal Technologist

- Sanitized **SQUINS** dataset with proper defect listing & classified metallic defects using **two-fold procedure** applied on the surface inspection system
- Implemented a **Convolutional Auto Encoder** to detect **27 types** of defects that would be further used to enhance the automation model

ACHIEVEMENTS

- Awarded certificate of recognition from **ITC Infotech** for giving the **best at work and contributing towards delivering exceptional results**.
- Won **Global Championship for PwC Digital - D'Art of Disruption, 2023** for **Cybersecurity & Data Privacy with Generative AI**.
- Secured **2nd position** for **Loyola School in Provoke, 2015**, a **hacking inter-school annual IT fest**.
- Secured **3rd position** for **Loyola School in Quantum, 2016**, a **web-design inter-school competition**.
- Selected as **National Student of the Week from Camp - K12**.
- Recognized as **Grand Finalist from Google for Sahana Software Foundation**, a humanitarian and non profit organisation in **Google Code-in, 2014** amongst **24 other open-source organizations in the world**.

RELEVANT PROJECTS

SAGE: Vocabulary Improving Game

- Developed **SAGE**, a multiplayer game, using **Angular, TypeScript, Firebase & Material Design**, for teachers & students wanting to improve vocabulary
- Designed for a **classroom setting**, students try to guess the meaning of a word through hints which is controlled by the teacher authenticated via **OAuth 2.0**

Point of Sales: Sales Transaction Application

- Developed API endpoints using **Angular, Node & MariaDB** having login utility, editing & selling a transaction for a specific user
- Login utility always **(re)loaded** on display screen showing username & password for a particular user & the final receipt could be **emailed** to the user

Retriever: Retrieving Information from Document

- Pre-processed the document data by **tokenization, stop word removal & stemming** in **Python** thus calculating the **term & document frequency**
- Frequencies were used to take out the **term score** & information was then extracted by taking out the **cosine similarity**

Samadhaan: Concern Logging Interface

- Build a concern logging application using **PHP, HTML, CSS & JS** where concerns can be logged by a user & an administrator can attend those concerns
- Final system was used in **Ambuja Cement Colony** at **Sindri, Jharkhand** who uses this application internally to log rental & house concerns

Axxelerate: Python based Search Engine

- Created a **Python** based search engine using **Scrapy** for crawling webpages & stores the data in **MariaDB** which indexed a **Binary Tree** with **Outer Joins**
- Implemented page ordering using **PageRank** algorithm & **Flask** for the server with **Angular & Material Design** approach for front end

Lexical Anazlyser: Finite State Machine Implementation

- Used **finite state machine** having a set of states, a set of **transitions**, & a string of **input data**
- Lexical analyzer was then implemented in **C** to recognize a list of **identifiers & non-negative integers**

RESEARCH PUBLICATION

- Journal: **"Breast cancer data classification using deep neural network"**, **IJISDC**, Volume 3, Number 2, 2021, Article 115169

CERTIFICATIONS

- **AWS: Certified Solutions Architect**
- **AWS: APN Partner Cloud Economics**
- **AWS: APN Partner Business Professional**
- **AWS: Certified Cloud Practitioner**
- **Coursera: Python Development on AWS**
- **Coursera: Object Oriented Design Course**

TECHNICAL SKILLS

Programming Languages	Python, C, Java, ES6, PHP, HTML, SCSS, SQL, NoSQL, R, YAML
Tools & Frameworks	Flask, SpringBoot, Single-Spa, Webpack, Node, React, Vue, Angular, Next, Pytest, JUnit, Llamaindex, Langchain
Software & Services	AWS, Power Portals, GitHub, Azure DevOps, Jenkins, TravisCI, Docker, Kubernetes, PowerBI, Alteryx
Data ETL-Analysis & DBMS	Pandas, Keras, Numpy, SkLearn, Tensorflow, Spark, MS-SQL, MongoDB, Maria DB, DynamoDB, Redshift, Dataverse

POSITIONS OF RESPONSIBILITY

- Teaching Assistant
 - Research Assistant
- Digital Media Computation using Python, UMM, USA

Power Consumption Controlled Analysis, UMM, USA
- May 2017 - July 2017

Nov 2017 - Mar 2018

RELEVANT COURSEWORK

Software Design Patterns	Foundations of Computer Science	Probability & Statistics	Introduction to Databases
Object Oriented Design	Data Structures & Algorithms	Introduction to Machine Learning	Introduction to Operating Systems
Software Design & Development	Numerical Methods	Neural Networks & Deep Learning	Introduction to Compiler Design

FUN TAGS

Basketball, Football, CSGO, FIFA, Michael Jordan, Kobe Bryant, Goku, Vegeta, Burgers, Biryani, Shakes, Praying, Helping, Dancing, Gym, Nature & Singing